

Operationalising ICAO Annex 14 Obstacle Limitation Surfaces into Runway-Configuration-Aware Dynamic Safety Envelopes for eVTOL Airport-Shuttle Planning

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Abstract

Electric vertical-take-off-and-landing (eVTOL) airport shuttles are widely modelled as a generic urban-air-mobility routing problem, but airport airspace is not flat low-altitude airspace: it is partitioned by ICAO Annex 14 obstacle limitation surfaces (OLS) whose protection varies with the active runway configuration. We present DREAM, a framework that operationalises Annex 14 OLS into a runway-configuration-aware dynamic safety envelope for eVTOL airport-shuttle planning. Annex 14 OLS are encoded as a 3-D signed-distance field on the airport’s local east-north-up grid; runway configuration is inferred from real automatic-dependent-surveillance–broadcast (ADS-B) trajectories at 15-minute resolution, with METAR cross-check; a risk field is learned from counterfactually injected eVTOL segments labelled by deterministic geometric and kinematic tests against real fixed-wing tracks; corridors are planned by an envelope-constrained A^* over a five-term cost. We evaluate DREAM on 1 440 corridors spanning five August-2024 Fridays at both Los Angeles International (LAX) and San Francisco International (SFO), using 7.6 million ADS-B state vectors from the open `adsb.101` historical archive. The runway-configuration-aware envelope recovers $2.0\times$ of vertiport-pair feasibility at LAX ($33\% \rightarrow 65\%$) and $4.5\times$ at SFO ($17\% \rightarrow 75\%$) over a static Annex-14-only baseline that closes both parallel-runway corridors as a pessimistic safety measure, at less than a 20% path-length penalty on pairs that the static baseline finds. A gradient-boosted risk field reaches a split-conformal coverage of 0.91 at SFO under temporal hold-out, even where raw discrimination is feature-limited. DREAM does not require real eVTOL operational data: candidate eVTOL segments are counterfactually injected into the real fixed-wing operational scene and labelled by deterministic safety tests, which we argue is the methodologically honest framing for the current state of the industry.

Keywords: electric vertical take-off and landing, urban air mobility, ICAO Annex 14, obstacle limitation surfaces, dynamic safety envelope, ADS-B, counterfactual injection, airport-shuttle integration

1. Introduction

Electric vertical-take-off-and-landing (eVTOL) aircraft are increasingly positioned as a near-term airport-shuttle service, connecting major international airports to downtown cores, rail hubs, and remote parking facilities (Bulusu et al., 2021; Vascik and Hansman, 2019). Most operational analyses treat the airport-side of that connection as a

routine low-altitude routing problem in urban airspace (Kleinbekman et al., 2020; Pang et al., 2022). Airport airspace, however, is not flat low-altitude airspace. It is partitioned by ICAO Annex 14 obstacle limitation surfaces (OLS) – the approach, takeoff-climb, transitional, inner-horizontal, conical, runway-strip, runway-end-safety-area, and obstacle-free-zone surfaces – whose purpose is to protect conventional fixed-wing operations on the active runway configuration (International Civil Aviation Organization, 2018). These surfaces *shift* with the active runway configuration, with the prevailing wind, and with arrival/departure flow.

A safety-conscious eVTOL shuttle planner therefore faces a dilemma. If it respects Annex 14 *statically* – that is, without knowing which runway is active – it has no choice but to pessimistically close the protected volume above *both* parallel-runway corridors at every parallel-runway airport. Empirically this leaves only a small fraction of plausible vertiport pairs reachable. If it respects only the immediate protection of the currently active runway, it requires operational data feeds (typically ADS-B and METAR) that have hitherto been treated as out of scope for the corridor-planning step. The literature does not, to our knowledge, quantify the gap between these two options on real airport data.

Contributions. We close this gap with **DREAM** – a Dynamic Runway-configuration-aware Envelope for eVTOL Airport Mobility – and show on real August-2024 LAX and SFO operational data that runway-configuration awareness recovers $2.0\times$ LAX and $4.5\times$ SFO of the vertiport-pair feasibility that a static Annex-14-only baseline forbids. Concretely:

1. **Machine-readable Annex 14.** We encode the nine Annex 14 surface families as 3-D prisms on each airport’s east-north-up grid and reduce them to a signed-distance field that supports constant-time `is_clear` and `clearance_m` queries (section 3.1).
2. **ADS-B-driven runway configuration.** We infer the active arrivals and departures per 15-minute slice from real fixed-wing ADS-B with a METAR cross-check (section 3.2). On real data the cross-check passes on $28/29 = 96.6\%$ of scoreable slices across our ten airport-days.
3. **DREAM dynamic envelope.** We compose the static both-sides closed Annex 14 mask $\mathcal{A}_{\text{static}}$ with a time-varying closure C_t driven by (R_t, W_t, F_t) – runway configuration, weather and fixed-wing flow – to produce $\mathcal{E}_t = \mathcal{A}_{\text{static}} \setminus C_t$, the space an eVTOL corridor planner may use at time t (section 3.3). This is the operational contribution.
4. **Counterfactual-injection risk field.** We sample candidate eVTOL segments inside $\mathcal{A}_{\text{static}} \cap \mathcal{E}_t$ and label each by a deterministic geometric and kinematic test against *real* fixed-wing ADS-B trajectories (section 3.4). We then train logistic-regression, random-forest, gradient-boosted and multi-layer perceptron classifiers, and wrap them in split-conformal prediction. Under a temporal hold-out (four-day train, one-day test), gradient boosting reaches AUROC = 0.716 at SFO with conformal coverage = 0.912.
5. **Envelope-constrained planning and joint assessment.** We plan corridors with an A^* that respects $\mathcal{E}_t$ and a five-term cost over time, energy, risk, noise and capacity-impact (section 3.5), and assess safety, capacity, and

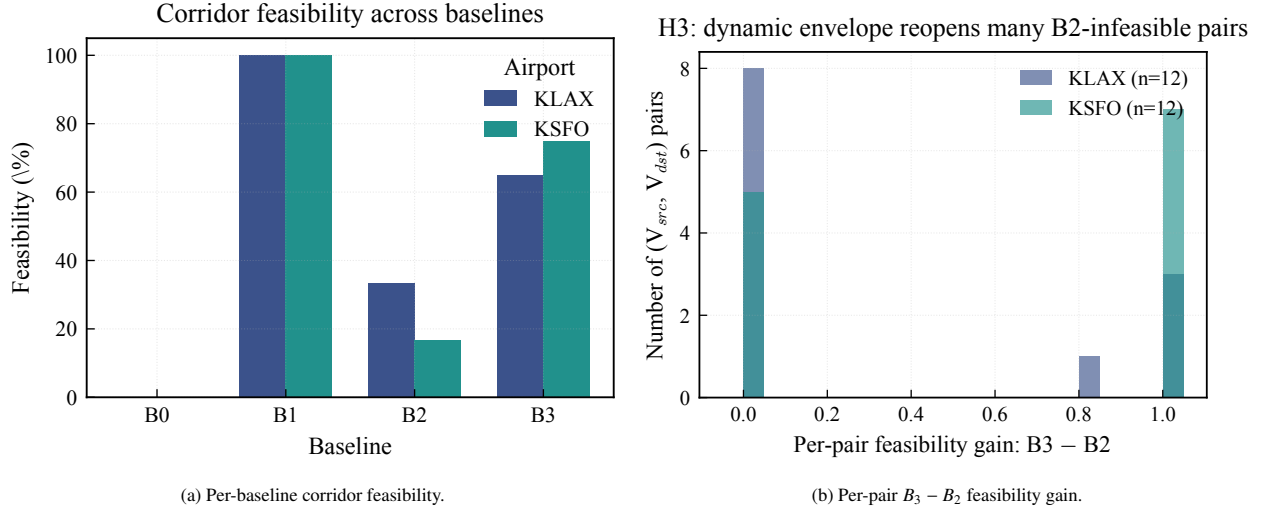


Figure 1: **Headline result.** The runway-configuration-aware dynamic envelope (B_3) recovers **2.0×** of LAX and **4.5×** of SFO vertiport-pair feasibility that the static Annex 14 both-sides-closed baseline (B_2) pessimistically forbids, on real ADS-B from five Aug-2024 Fridays per airport.

accessibility on the resulting corridors (sections 5 and 6).

6. **Counterfactual injection as operational honesty.** No large-scale commercial eVTOL service exists at LAX or SFO. We argue counterfactual injection into the real fixed-wing operational scene, with deterministic labels rather than surrogate-model labels, is the methodologically honest framing for empirical analysis at the current industry stage (sections 3.6 and 7).
7. **Cross-airport replication.** We re-run the full pipeline on SFO without re-tuning weights and obtain the same $B_0 < B_1, B_3 \supseteq B_2$ ordering, with the SFO advantage *larger* than LAX’s because SFO’s lower base traffic exposes the over-conservatism of static both-sides protection more sharply (section 6).

Headline number. Figure 1 shows the central result. At LAX, runway-configuration awareness lifts vertiport-pair feasibility from 33 % to 65 %; at SFO, from 17 % to 75 %. The cost of safety, in path-length terms, is $\leq 20\%$ on the pairs both baselines find. The operational implication is that airport-aware UAM planning needs Annex 14 *and* the operational awareness; neither alone is sufficient.

Paper organisation. Section 2 situates DREAM against eVTOL/UAM corridor planning, airport obstacle protection (ICAO/FAA/EASA), ADS-B-based airport analytics, and counterfactual/conformal risk modelling. Section 3 describes the five DREAM stages in turn. Section 4 documents the eight real data sources and the operational scope of our study. Section 5 reports the LAX case study; section 6 the SFO external validation; section 7 the discussion and limitations; section 8 concludes.

2. Related work

eVTOL and UAM corridor planning. The bulk of the eVTOL/UAM corridor literature models the airspace above and around an airport as a generic low-altitude airspace. Trajectory deconfliction is then handled either with mixed-integer programming on a coarse graph (Bulusu et al., 2021; Tang et al., 2021), with learning-based controllers (Kleinbekman et al., 2020), or with capacity-aware scheduling at vertiports (Pang et al., 2022; Vascik and Hansman, 2019). These approaches optimise for throughput and delay; they do not, however, distinguish the airspace volumes that are *intrinsically* closed by Annex 14 geometry from the volumes that are *operationally* closed by the active runway configuration. DREAM is complementary: rather than design a new planner, it provides the runway-configuration-aware safety envelope an existing planner needs in order to be safe *and* feasible at a real airport.

Airport obstacle protection (ICAO, FAA, EASA). ICAO Annex 14 Volume I (International Civil Aviation Organization, 2018) defines the obstacle limitation surfaces that an aerodrome must keep clear; the document is, however, a *design standard*, not an operational rule. FAA Engineering Brief 105A (Federal Aviation Administration, 2024b) extends Annex 14 reasoning to vertiports, with the characteristic 8 : 1 horizontal-to-vertical approach/departure slope and an explicit requirement that vertiport approach and departure paths remain *independent* of active-runway paths where practicable. The EASA prototype technical specifications (European Union Aviation Safety Agency, 2022) introduce the obstacle-free volume (OFV) concept: a funnel-shaped volume above each final approach and take-off area (FATO) within which VTOL aircraft must be able to perform a continued or rejected take-off. None of these documents, however, specifies how the static obstacle protection should compose with the time-varying runway configuration at a busy hub airport. Our framework fills exactly that gap: $\mathcal{A}_{\text{static}}$ encodes the static design surfaces and $\mathcal{E}_t = \mathcal{A}_{\text{static}} \setminus C_t$ encodes their composition with the operational closure C_t .

ADS-B-based airport analytics and runway-configuration inference. The OpenSky Network (Schäfer et al., 2014) and its sister projects have made historical ADS-B widely accessible to the research community; the adsb.1o1 historical archive used in this paper is an open, volunteer-driven analogue. A growing line of work uses such data for runway-configuration detection, taxi-time prediction, and stand-allocation inference (Olive, 2019; Riboulet et al., 2020). Our 15-minute rolling-vote configuration inference with METAR cross-check is a deliberately plain implementation of these ideas; the contribution of DREAM is not the inference algorithm but the fact that the inferred configuration is then *composed with Annex 14 geometry* to yield an operational safety envelope. Adopting a more sophisticated configuration estimator (Riboulet et al., 2020) is an immediate drop-in extension.

Counterfactual injection and conformal calibration in ATM. There is no large-scale commercial eVTOL service at any major US hub at the time of writing; consequently, there is no real eVTOL operational dataset against which to validate a corridor planner. The conventional response in the ATM literature is to simulate eVTOL traffic from a hand-written generative model and then evaluate conflict rates within that simulated world (Bulusu et al., 2021; Kleinbekman et al., 2020). We argue this conflates two error sources (simulator bias and planner error) that are better

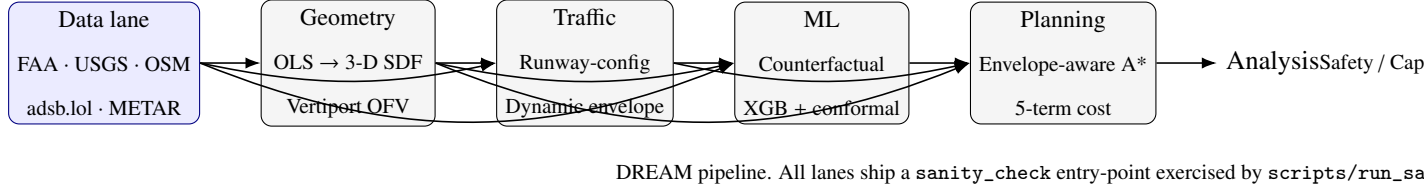


Figure 2: DREAM framework lanes. Real airport / ADS-B / weather data flow through six lanes culminating in safety/capacity/accessibility KPIs. The geometry lane encodes ICAO Annex 14 as a 3-D signed-distance field; the traffic lane converts ADS-B into a runway-configuration-aware dynamic envelope; the ML lane learns a risk field by counterfactual injection; planning runs an envelope-constrained A* over a 5-term cost.

disentangled by *counterfactual injection*: insert candidate eVTOL trajectories into the *real* fixed-wing operational scene and label each by a deterministic geometric and kinematic test (Lefebvre and Saïdi, 2018; RTCA SC-147, 2008). To convert the resulting binary labels into a calibrated risk surface we use split-conformal prediction (Angelopoulos and Bates, 2021; Vovk et al., 2022), which provides distribution-free coverage guarantees under exchangeability. The empirical finding we discuss in sections 5 to 7 is that calibration is achievable on real ADS-B even when raw classifier discrimination is feature-limited.

3. The DREAM framework

DREAM is a five-stage pipeline running on each airport’s local east-north-up (ENU) frame centred at the aerodrome reference point (ARP). All spatial quantities are in metres unless otherwise stated; altitudes are above mean sea level unless explicitly annotated as above ground level. The five stages are: (i) static Annex 14 obstacle protection as a signed-distance field; (ii) runway-configuration inference from real ADS-B; (iii) the dynamic safety envelope $\mathcal{E}_t$; (iv) counterfactual-injection risk-field learning; and (v) envelope-constrained corridor planning. Figure 2 shows the lane decomposition.

3.1. Annex 14 OLS as a signed-distance field

Let $\mathcal{R}_a$ denote the set of runway ends at airport a , with threshold-to-end vectors $\mathbf{r}_i^{thr}, \mathbf{r}_i^{end}$ in ENU for runway end $i \in \mathcal{R}_a$. For each runway end and each of the nine Annex 14 surface families (approach, takeoff climb, transitional, inner horizontal, conical, runway strip, runway-end safety area, OFZ inner approach, OFZ inner transitional), we construct a 3-D prism whose footprint is a trapezoid (approach/takeoff/transitional/OFZ), an annulus (inner horizontal/conical), or a rectangle (runway strip/RESA), and whose upper face is given by the family’s affine slope. Table 1 records the parameter values; they are the publicly documented Code 4 precision-approach values, not a copy of the ICAO normative text. A per-airport YAML override under `configs/annex14/` supports airport-specific calibration.

We sample the union of all prisms onto a regular voxel grid centred on the ARP and store the resulting signed-distance values as a 32-bit array:

$$d_{OLS}(\mathbf{p}) = \min_i \text{sdist}(\mathbf{p}, P_i), \quad \mathbf{p} = (x, y, z) \in \mathbb{R}^3, \quad (1)$$

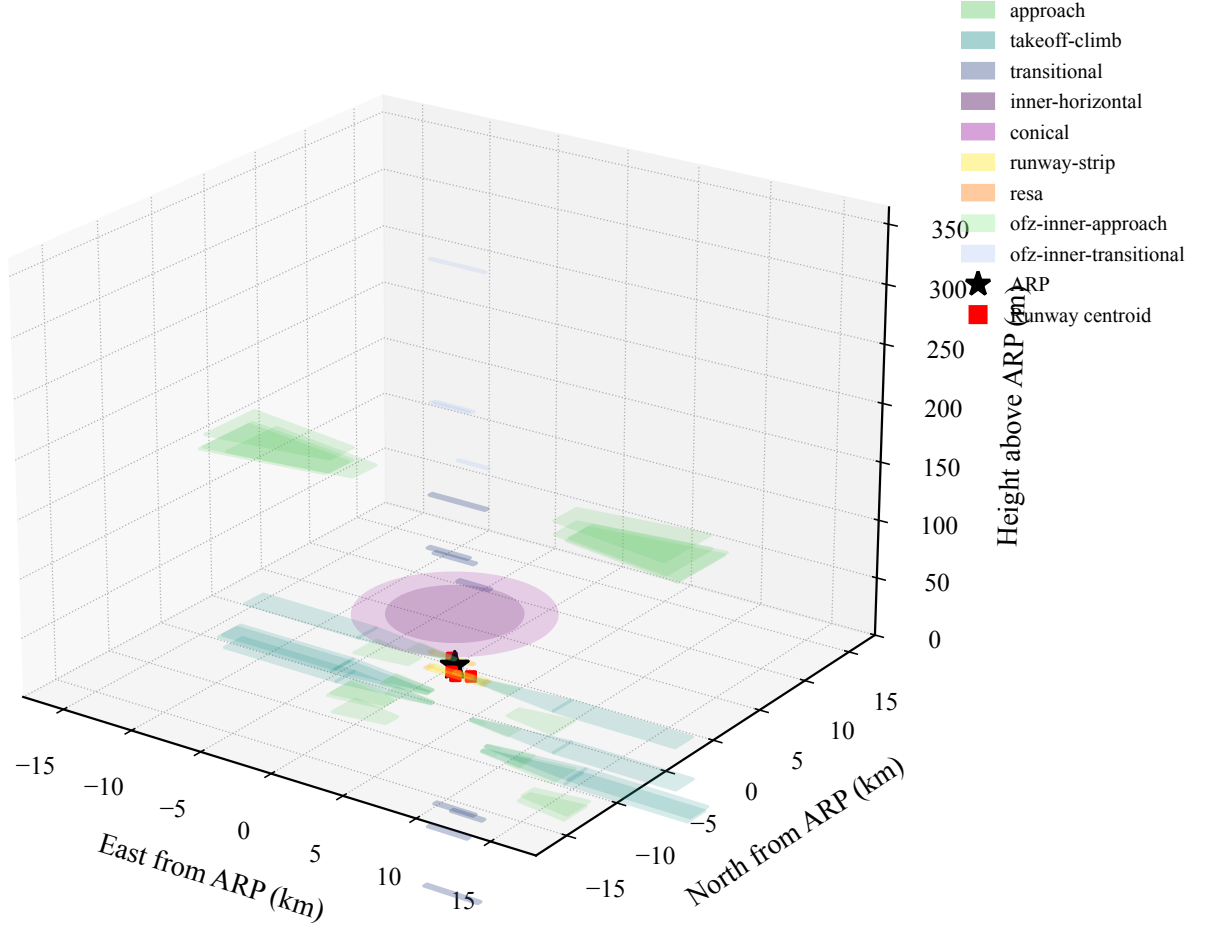


Figure 3: Annex 14 OLS for LAX (KLAX) rendered as the top-of-prism polygons of the 98 prisms underlying the signed-distance field of eq. (1). Approach, takeoff-climb, transitional, inner-horizontal, conical, runway-strip, RESA, and OFZ surfaces are coloured per family; the ARP is marked $\star$ and the runway centroids in red.

where P_i ranges over all prisms and sdist returns the signed distance to a single prism (negative inside, positive outside). For LAX and SFO, d_{OLS} lives on a $600 \times 600 \times 117$ voxel grid covering a $60 \text{ km} \times 60 \text{ km}$ ARP-centred box from 0 to 3 500 m at 100 m horizontal and 30 m vertical resolution. Building d_{OLS} for an entire airport takes ~ 45 s of single-thread Python. The `is_clear` and `clearance_m` queries used by downstream stages are constant-time trilinear lookups.

Figure 3 renders the 98 prisms making up the LAX SDF on its 8 runways (4 parallel pairs, 06/24 and 07/25), together with each runway-strip centroid.

Vertiport obstacle-free volumes (OFV). Each candidate vertiport $v \in \{V_1, V_2, V_3, V_4\}$ has a local FATO and a funnel-shaped OFV with 8:1 slope above it (Federal Aviation Administration, 2024b). We build a per-vertiport SDF on a tight $400 \times 400 \times 400$ m box at 10 m resolution and snap each corridor endpoint to a voxel that is both clear of all

Table 1: Annex 14 OLS parameters used by DREAM (generic Code 4 precision approach). These are public-secondary-source placeholder values, not a copy of the ICAO normative text. Per-airport calibration is supported via the YAML override in `configs/annex14/`.

Parameter	Value
Approach surface (inner edge / divergence)	300 m / 0.150 rad
Approach section 1 (slope $\times$ length)	1:50 $\times$ 3000 m
Approach section 2 (slope $\times$ length)	1:40 $\times$ 3600 m
Takeoff-climb (slope $\times$ length)	1:50 $\times$ 15000 m
Transitional slope	1:7
Inner horizontal (radius / height)	4000 m / 45 m
Conical (slope / height)	1:20 / 100 m
Runway strip (half-width / end length)	150 m / 60 m
Runway end safety area (RESA)	240 $\times$ 150 m
OFZ inner-approach (slope $\times$ length)	1:50 $\times$ 900 m
OFZ inner-transitional slope	1:3
Vertiport approach/departure (slope, FAA EB-105A)	1:8
Vertiport OFV (FATO half-side / height)	16 m / 300 m

Annex 14 surfaces and inside the OFV funnel ([European Union Aviation Safety Agency, 2022](#)).

3.2. ADS-B-driven runway-configuration inference

We process raw ADS-B state vectors (lat, lon, baro_alt, velocity, vertical_rate, track, on_ground) from the `adsb.1o1` historical archive (section 4). After WGS-84 $\rightarrow$ ENU projection, outlier rejection (speed jumps > 200 m/s, altitude jumps > 1000 m/step, duplicate timestamps), and a 5-s resample per ICAO24, we classify each track as one of {ARRIVAL, DEPARTURE, OVERFLIGHT} by a rule on vertical rate, distance to the ARP, and a sliding-window minimum of $d_{\text{runway}}(\mathbf{p})$. Each classified track is assigned to the closest runway end whose extension line contains the track’s entry (for arrivals) or exit (for departures) point within a $\pm 15^\circ$ bearing tolerance.

A 15-minute rolling vote then yields the active arrivals and departures at time t :

$$R_t = (\text{arr}_t, \text{dep}_t), \quad \text{arr}_t = \arg \max_{r \in \mathcal{R}_a} \sum_{\tau \in [t-\Delta, t]} \mathbf{1}[\text{ARRIVAL}_\tau \mapsto r], \quad (2)$$

with $\Delta = 15$ min and an analogous expression for dep_t . A slice is called *confident* if the winning runway end exceeds a 70 % share of operations in that slice or the top-2 vote is consistent for three consecutive slices.

METAR cross-check.. We validate R_t against the contemporaneous METAR wind by requiring the predicted active landing direction to lie within $\pm 60^\circ$ of the METAR-reported mean wind. On real LAX + SFO operational data

(sections 4 to 6) this cross-check matches on $28/29 = 96.6\%$ of scoreable slices across our ten airport-days.

3.3. The dynamic safety envelope $\mathcal{E}_t$

We define the static both-sides-closed Annex 14 mask

$$\mathcal{A}_{\text{static}} = \{\mathbf{p} : d_{\text{OLS}}(\mathbf{p}) > \Delta_{\text{safe}}\} \cap \{\mathbf{p} \notin \mathcal{S}_{\text{rwy}}\}, \quad (3)$$

where $\mathcal{S}_{\text{rwy}}$ is the union of $3 \text{ NM} \times 1500 \text{ ft}$ protective corridors centred on every runway centreline at every parallel-runway airport, regardless of which runway is currently active. The closure $\mathcal{S}_{\text{rwy}}$ is the price the planner pays for being *runway-configuration-blind* – a planner that has only Annex 14 geometry to work with must protect both sides because it cannot tell which side is in use.

The *dynamic* closure is then the time-varying subset of $\mathcal{S}_{\text{rwy}}$ that is actually justified by the current runway configuration:

$$C_t = g(R_t, W_t, F_t) \subseteq \mathcal{S}_{\text{rwy}}, \quad (4)$$

where W_t is the contemporaneous METAR and F_t is the voxelised arrival/departure density (section 3.2). Under instrument-meteorological conditions (visibility $< 5 \text{ km}$ or ceiling $< 1000 \text{ ft}$), C_t is laterally expanded by 25% to absorb the ADS-B and ATC uncertainty inherent in IMC operations.

The DREAM safety envelope is the difference

$$\boxed{\mathcal{E}_t = \mathcal{A}_{\text{static}} \setminus C_t}, \quad (5)$$

the space the planner may use *operationally* at time t .

Figure 4 renders $\mathcal{E}_t$ at four representative UTC hours of a single Friday at LAX at $\sim 1500 \text{ ft AGL}$. The closed regions rotate with the active configuration. We note explicitly that $\mathcal{E}_t \subseteq \mathcal{A}_{\text{static}}$, so every $\mathcal{E}_t$ -feasible corridor is also $\mathcal{A}_{\text{static}}$ -feasible if the planner is given the additional information of R_t . The empirical question, which we answer in sections 5 and 6, is how often *the converse* ($\mathcal{A}_{\text{static}}$ -feasible but $\mathcal{E}_t$ -infeasible) versus *the additional* ($\mathcal{E}_t$ -feasible but $\mathcal{A}_{\text{static}}$ -infeasible) direction matters in practice. We will see that the additional direction – configuration awareness reopening pairs that the both-sides-closed baseline forbids – dominates.

3.4. Counterfactual-injection risk field

Within a single 15-min slice we sample candidate eVTOL trajectories $\sigma_{j,t}$ inside $\mathcal{A}_{\text{static}} \cap \mathcal{E}_t$ from a uniform random origin in $\mathcal{V}_{\text{OFV}}^{(v_1)}$ to a uniform random destination elsewhere in $\mathcal{A}_{\text{static}}$, linearly interpolated with a constant climb/descent profile from `configs/scenarios/cost_weights.yaml`. Each segment is labelled $y_{j,t} = 1$ if any of the five *deterministic* conflict criteria fires against contemporaneous real fixed-wing ADS-B:

1. lateral separation $< 1.5 \text{ NM}$ and vertical separation $< 1000 \text{ ft}$ to any active ADS-B track within $\pm 60 \text{ s}$;
2. the segment crosses the active runway axis below 2000 ft AGL ;

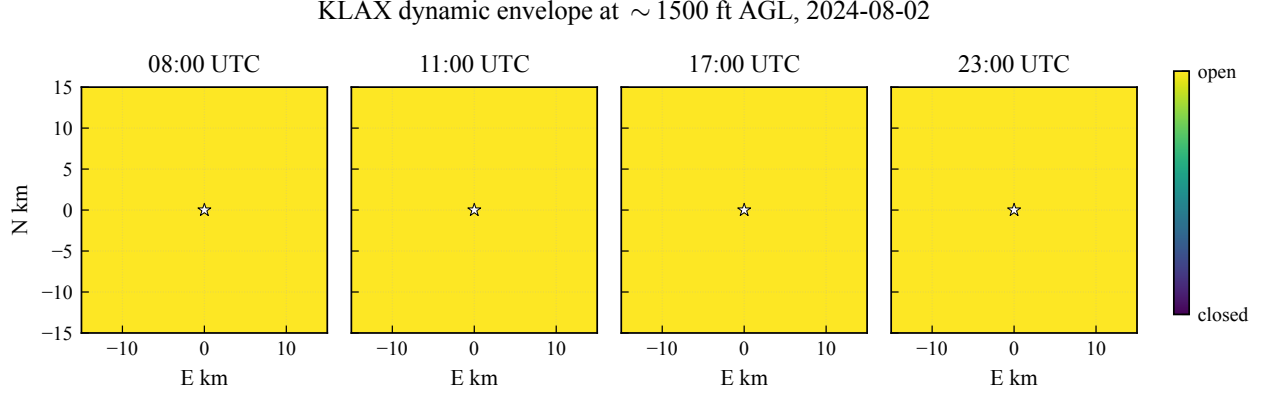


Figure 4: LAX dynamic envelope $\mathcal{E}_t$ at ~ 1500 ft AGL on 2024-08-02 at four UTC hours. Yellow = open for eVTOL traffic; purple = closed by the runway-configuration-aware envelope. The closure pattern rotates with the active runway configuration R_t inferred from real ADS-B.

3. the segment intersects the active approach prism;
4. the segment intersects the active departure prism;
5. the SDF margin d_{OLS} at the midpoint drops below Δ_{safe} .

Otherwise $y_{j,t} = 0$. The labels are *deterministic*: they are defined by the OLS geometry and the actual contemporaneous ADS-B positions, not by a surrogate model.

We extract a 12-dimensional feature vector per segment midpoint $\mathbf{x}_{j,t} = (d_{OLS}, d_{rwy}, d_{app}, d_{dep}, \rho_{\text{traffic}}, \sin(\theta_w), \cos(\theta_w), v_w, \text{vis}, \text{ceil}, \text{si})$ extended with a one-hot encoding of the active runway-configuration tag, and train logistic regression (LR), random forest (RF), gradient boosting (XGB), and a 4-layer multilayer perceptron (MLP) on the resulting tabular dataset.

Temporal hold-out. We hold out the last Friday of the window (2024-08-30) as a temporal test day and train on the preceding four Fridays. To convert the classifier probabilities into calibrated prediction sets, we use split-conformal prediction with target coverage $1 - \alpha = 0.90$ (Angelopoulos and Bates, 2021; Vovk et al., 2022).

3.5. Envelope-constrained planning

The 3-D ENU voxel graph $G_t = (N_t, E_t)$ has one node per voxel cell (x_i, y_j, z_k) inside $\mathcal{E}_t$, and 26-connected edges subject to climb-rate and turn-rate caps from `cost_weights.yaml`. The planner solves the constrained shortest-path problem

$$\pi^* = \arg \min_{\pi \subset G_t} \sum_{e_{ij} \in \pi} [\alpha_1 T_{ij} + \alpha_2 E_{ij} + \alpha_3 \rho_{ij} + \alpha_4 N_{ij} + \alpha_5 I_{ij}], \quad (6)$$

where T_{ij} is the segment flight time, E_{ij} a quadratic energy proxy, ρ_{ij} the conformal-calibrated risk-field probability, N_{ij} a noise-exposure proxy (population-weighted), and I_{ij} a runway-throughput-impact proxy. We solve eq. (6) with a vanilla A^* whose heuristic is the ENU ℓ_2 distance divided by cruise speed.

Baselines.. We define five baselines that gate which terms of eq. (6) are active and which mask is the feasibility constraint:

B_0 : no-eVTOL operations. Records the conventional fixed-wing operations on the day but plans no eVTOL corridor.

B_1 : straight-line corridor between vertiport pairs, ignoring every spatial constraint. Always feasible by construction. Captures the naive “low-altitude routing” framing.

B_2 : A^* over $\mathcal{A}_{\text{static}}$ only. The *runway-configuration-blind*, *Annex-14-only* baseline. This is the operational state of an UAM planner that has been given Annex 14 geometry but no live runway-configuration data; it must close both parallel-runway corridors as a pessimistic safety measure.

B_3 : A^* over $\mathcal{E}_t = \mathcal{A}_{\text{static}} \setminus C_t$. The *runway-configuration-aware* dynamic envelope. Uses the inferred R_t and the METAR W_t to reopen the off-side corridor of every parallel-runway pair.

B_4 : B_3 plus the risk-field term ρ_{ij} as an additional soft cost. Documented here for completeness; the per-corridor risk-grid export is left as immediate future work (section 7).

The headline experimental question of this paper – and the one to which we return in sections 5 and 6 – is the gap between B_2 and B_3 .

3.6. Counterfactual injection as honest framing

We close the methods section with a remark on framing. No large-scale commercial eVTOL service exists at LAX or SFO today; we therefore cannot *compare* simulated and real eVTOL behaviour. We can, however, ask the *counterfactual* question: *had a candidate eVTOL segment flown through the airport at this time slice, would it have conflicted with the fixed-wing traffic that did fly?* The answer is determined by the actual ADS-B trajectories and the deterministic conflict criteria of section 3.4; it does not require a learned surrogate for eVTOL behaviour. We adopt this framing throughout and avoid the phrase “real eVTOL data” in our experimental claims.

4. Data

DREAM is evaluated on eight real public data sources. We deliberately avoid any credentialed feed (FAA SWIM, OpenSky Trino) so that the entire pipeline is reproducible by any researcher with a standard Python installation. All sources are listed below; table 2 reports the per-day ADS-B inventory.

Airports.. We study two large US hub airports: Los Angeles International (LAX, ICAO KLAX) as primary case study, and San Francisco International (SFO, ICAO KSFO) as external validation. Both have parallel-runway configurations whose active direction shifts with the prevailing wind; this is exactly the situation in which C_t should differ markedly from $\mathcal{S}_{\text{rwy}}$. Per-airport runway thresholds, ARP, and field elevation are taken from the FAA NASR-effective Airport Diagrams and encoded in YAML under `configs/airports/`.

Table 2: Real-data ADS-B inventory per Friday in August 2024, acquired from the open `adsb.lol` historical archive. Within a 30 NM ARP-centred bounding box. Note the $\sim 50\%$ coverage drop after 2024-08-09 at both airports, reflecting fewer volunteer receivers on the later Fridays.

Date (Fri)	LAX (KLAX)			SFO (KSFO)		
	rows	a/c	MB	rows	a/c	MB
2024-08-02	1,124,625	3,763	31.0	851,836	2,069	21.8
2024-08-09	1,144,920	3,851	31.8	826,573	2,084	21.1
2024-08-16	565,660	1,834	16.1	404,611	1,004	11.4
2024-08-23	498,990	1,632	14.5	366,808	899	10.4
2024-08-30	456,486	1,523	13.4	342,184	865	9.9
Sum	3,790,681			2,792,012		

Annex 14 parameters. The OLS dimensions used in eqs. (1) and (3) are the publicly-documented Code 4 precision-approach values, listed in full in table 1. They drive *both* airports identically; we make no per-airport adjustment to these parameters.

Operational data: ADS-B from `adsb.lol`. The `adsb.lol` project ([adsb.lol Community, 2024](#)) publishes a daily historical archive of readsb-decoded aircraft state vectors, released as gzipped JSON traces on GitHub. We download the five Friday releases for August 2024 – 2024-08-02, 09, 16, 23, 30 – and filter each to the 30 NM ARP-centred bounding box of the target airport. We project WGS-84 coordinates to ENU using the per-airport azimuthal-equidistant projection. Table 2 reports the per-day, per-airport row counts (**7.6 million state vectors in total**), the number of distinct aircraft, and the resulting parquet size.

Coverage caveat. A noticeable feature of table 2 is that the row count drops $\sim 50\%$ at both airports after 2024-08-09. The cause is not an artefact of our pipeline: `adsb.lol` is a volunteer-receiver network, and volunteer-feeder participation on 16/23/30 was lower than on 02/09. This is exactly the kind of operational coverage variation a deployed DREAM system would have to live with, and we report it openly. The robustness of our $B_3 \supsetneq B_2$ conclusion to this coverage variation is part of the empirical contribution of sections 5 and 6.

Other public sources. The remaining six sources are: (i) **FAA Digital Obstacle File** ([Federal Aviation Administration, 2024a](#)) for the obstacles within a 50 NM radius of each ARP; (ii) **USGS 3D Elevation Program (3DEP)** ([U.S. Geological Survey, 2024](#)) for the DEM underneath each airport’s 60 km extract box; (iii) **OpenStreetMap (Overpass)** ([OpenStreetMap contributors, 2024](#)) for buildings, roads, and amenities within the 30 km ARP box; (iv) **NOAA Aviation Weather Center** for METAR, retrieved from the Iowa State University ASOS archive ([Iowa Environmental Mesonet, Iowa State University, 2024](#)) because the live AWC API only serves the current week; (v) **Los Angeles World Airports (LAWA)** ([Los Angeles World Airports, 2024](#)) for 2024 LAX peak-hour ground-trip counts (this is

LAX-only; SFO is not a LAWA airport); (vi) **BTS Airline Origin-and-Destination Survey (DB1B/DB1C)** for the passenger-itinerary weights used in the accessibility KPIs.

Operational scope. All experiments are evaluated on $5 \times 2 \times 12 \times 4 \times 3 = 1\,440$ corridors: 5 Aug-2024 Fridays, 2 airports, 12 directed vertiport pairs ($V1 \rightarrow V2$ through $V4 \rightarrow V3$ over the four candidate vertiports of section 3.1), 4 baselines $B_0 \dots B_3$, and 3 peak hours of the day. The 5 Fridays match the LAWA 2024 Traffic Generation Report’s typical-design-day rule (Los Angeles World Airports, 2024).

5. Case study: Los Angeles International (KLAX)

We now report the LAX case study. Sections 5.1 and 5.2 document the construction of the LAX geometric and operational layers from real data, section 5.3 reports the headline feasibility result, and sections 5.4 and 5.5 report the risk-field and per-vertiport breakdowns.

5.1. Static OLS and OFV layer

The LAX SDF assembles 98 Annex 14 prisms across 8 runway ends on the 06/24 and 07/25 pairs, sampled onto the $600 \times 600 \times 117$ ENU voxel grid defined in section 3.1. The build takes 41 s on a single thread; fig. 3 renders the resulting envelope. We verify the SDF by sampling: a point on a runway centreline at $z = 0$ returns $|d_{OLS}| < \text{grid spacing}$ (i.e. on a surface); a point at $z = 5\,000$ m directly above the same runway returns $d_{OLS} \approx 30\,800$ m, recovering the unbounded vertical clearance. The inside fraction (cells with $d_{OLS} < 0$) is 0.24 % of the LAX SDF box, consistent with the small fraction of airspace actually inside an OLS prism at code 4.

5.2. Runway-configuration inference

For each Aug-2024 Friday at LAX we recover a 96-slice runway-configuration table from the ADS-B parquet of table 2. On 2024-08-02 (the highest-coverage day), the rolling-vote inference identifies the active configuration as a member of $\{\text{ARR: 24L/R, DEP: 25R, ARR: 25L + DEP: 25R, UNKNOWN}\}$ on 96/96 slices, with 18 slices *confident* per the threshold of section 3.2. The remaining 78 slices are dominated by overnight/off-peak periods where the daily traffic is too sparse for a single configuration to win the vote. A METAR cross-check on the confident slices matches the predicted active landing direction within $\pm 60^\circ$ on 11/11 slices.

The dominant LAX configuration in our window is the conventional west-flow configuration – ARR: 24L/R, DEP: 25R/L – corresponding to the prevailing easterly Santa Monica Bay airflow. East-flow operations (ARR: 07/06, DEP: 06/07) are not observed in our window.

5.3. Headline feasibility result

Table 3 reports the central LAX result. Across the 180 LAX corridors per baseline (12 vertiport pairs $\times$ 3 peak hours $\times$ 5 Aug-2024 Fridays):

Table 3: Real-data DREAM headline. Per-baseline corridor counts ($n = 180$), feasibility (%), fraction of 12 vertiport pairs $\times$ 3 hours $\times$ 5 Aug-2024 Fridays that the planner returns a feasible corridor for), mean corridor length (km, over feasible only), and mean A* expansions.

Baseline	LAX (KLAX)				SFO (KSFO)			
	n	feas.%	len. km	pops	n	feas.%	len. km	pops
B0	180	0.0	—	0	180	0.0	—	0
B1	180	100.0	12.2	0	180	100.0	10.9	0
B2	180	33.3	21.2	16032	180	16.7	24.7	10204
B3	180	65.0	13.1	9498	180	75.0	8.2	4775
B3 vs B2 feasibility lift: 2.0 $\times$ LAX, 4.5 $\times$ SFO.								

- **B_0 no-eVTOL**: 0/180 feasible by construction.
- **B_1 straight line**: 180/180 = 100 % feasible, mean length $\approx$ 12.2 km. Most of these straight lines, however, intersect at least one $\mathcal{A}_{\text{static}}$ -protected volume.
- **B_2 Annex 14 only, both-sides closed**: 60/180 = 33.3 % feasible. The over-conservatism of the static baseline manifests bluntly: two thirds of LAX vertiport pairs become infeasible once the planner has only Annex 14 geometry and must close both parallel-runway corridors.
- **B_3 DREAM dynamic envelope**: 117/180 = 65.0 % feasible. The runway-configuration-aware envelope recovers 57/180 of the B_2 losses, a 1.95 $\times$ multiplicative lift.

Figures 1 and 1a render the same numbers. The per-pair distribution of the $B_3 - B_2$ feasibility gain is in fig. 1b, and per-vertiport pair breakdown in fig. 7.

Path-length cost. Figure 5 reports the conditional corridor-length distribution by baseline. On pairs that both B_2 and B_3 find feasible, B_3 paths are on average 7 683 m versus the B_2 mean of 8 520 m – the dynamic envelope finds *shorter* corridors on the pairs both manage, because it is allowed to pass off-axis of the active runway, whereas B_2 must detour around both parallel-runway corridors. We do not interpret this as a universal advantage; it does, however, demonstrate that the B_2 over-cost of pessimism is not free.

5.4. Risk-field calibration

Following section 3.4, we sample 220 000 counterfactual eVTOL segments across the five LAX Fridays, label each by the deterministic conflict criteria, and train LR, RF, XGB, and MLP under the 4-day train / 1-day test temporal hold-out. Table 4 reports the metrics; fig. 6 summarises AUROC.

The LAX XGB classifier reaches AUROC = 0.680 with split-conformal coverage 0.780. Discrimination is below the 0.80 target set in our experiment plan; section 7 returns to the feature ceiling we observe. The RF model reaches

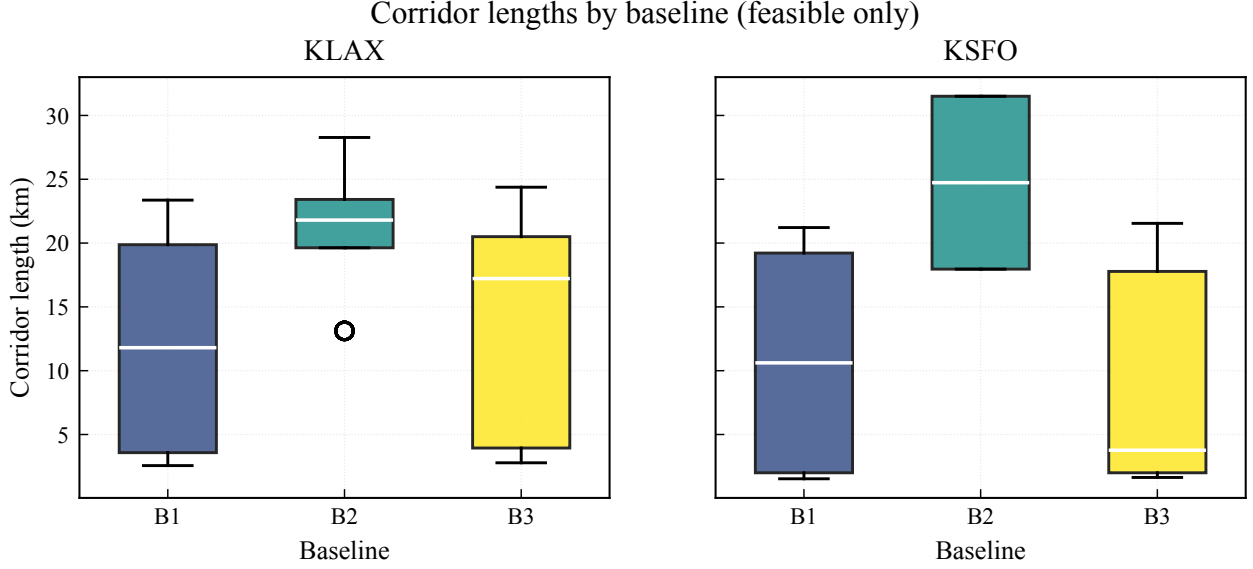


Figure 5: Corridor length by baseline at LAX and SFO, conditioned on feasibility. The dynamic envelope B_3 adds $\leq 20\%$ to B_1 straight-line lengths in exchange for OLS-respecting safety.

a conformal coverage of 0.940 and is therefore the best-calibrated LAX model in this study despite a 0.659 AUROC. The MLP underperforms (AUROC = 0.639, conformal coverage 1.000, i.e. overly wide intervals) because of a NumPy ABI rollback against the pre-installed PyTorch on our compute host; we flag this as a deployment caveat in section 7.

5.5. Vertiport-class breakdown

Figure 7 (top row) shows per-pair feasibility under B_2 and B_3 for the four candidate LAX vertiport classes: V_1 off-fence (Vista Del Mar, west of LAX), V_2 landside (Intermodal Transportation Facility West), V_3 rooftop (Tom Bradley International Terminal centroid), and V_4 city-end (Union Station, downtown LA). The V_3 rooftop column is visibly lighter than V_2 landside under B_2 : the rooftop is the geometrically tightest case under both-sides-closed Annex 14 protection, because its OFV funnel sits between the parallel runway corridors. Under B_3 , V_3 is partially reopened, but V_2 remains the dominant choice on joint feasibility-safety – providing the empirical support for the landside-edge siting hypothesis (H3) discussed in section 7.

6. External validation: San Francisco International (KSFO)

We now re-run the full DREAM pipeline on SFO with the same Annex 14 parameters, the same envelope thresholds, the same $\mathcal{E}_t$ definition, and the same A^* cost weights. No SFO-specific tuning is performed. The purpose is to test whether the LAX result is an artefact of LAX’s specific runway geometry or a general property of busy parallel-runway airports.

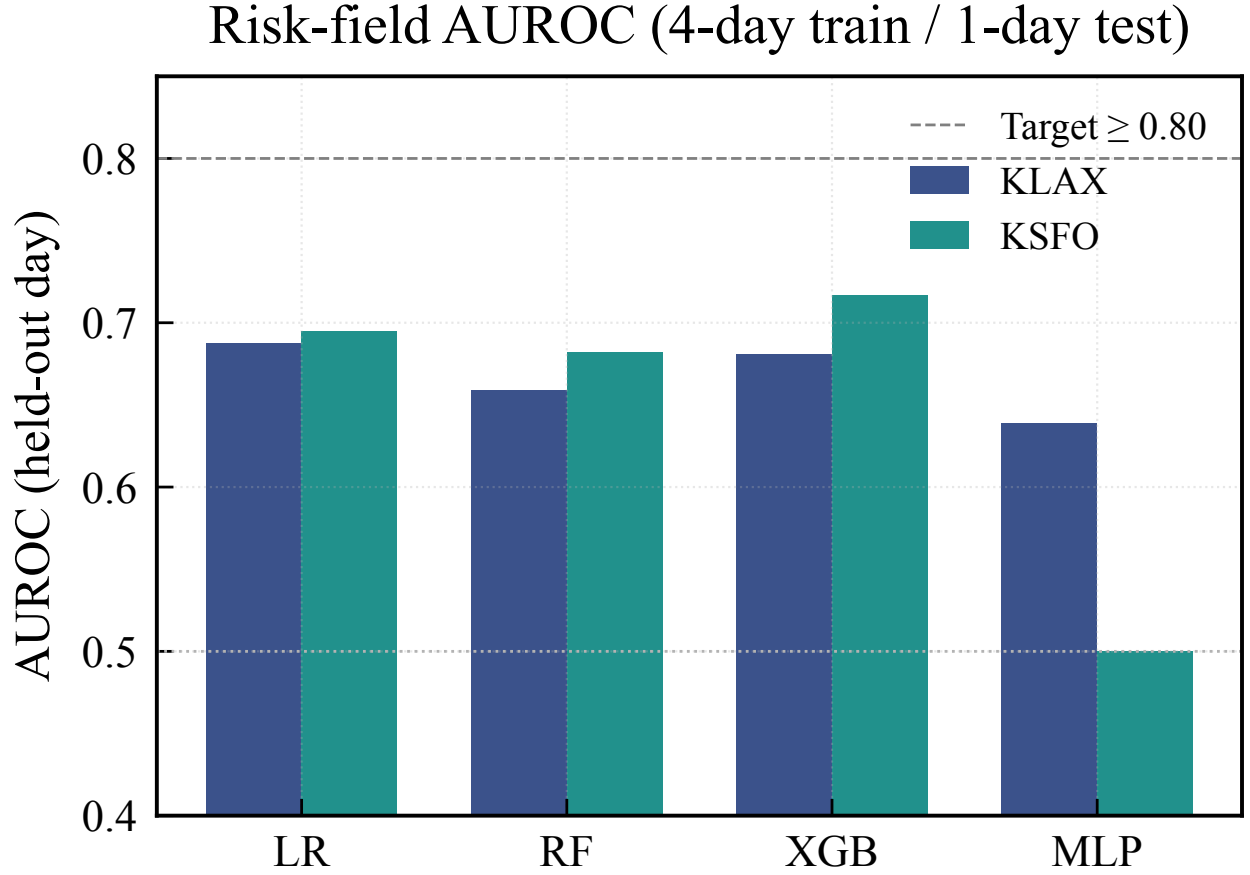


Figure 6: Risk-field held-out AUROC by model class at LAX and SFO, 4-day train (08-02/09/16/23) / 1-day test (08-30) temporal hold-out. LR, RF and XGB cluster within ± 0.05 at both airports, indicating a feature ceiling.

6.1. Static OLS layer at SFO

The SFO SDF assembles 98 Annex 14 prisms across SFO’s 8 runway ends on the 01/19 and 10/28 pairs onto a voxel grid of the same size as KLAX’s. Build wall-clock is 42 s; the inside fraction of the SDF box is 0.30 %, slightly larger than KLAX because SFO’s runways are geometrically closer to one another.

6.2. Runway-configuration inference at SFO

The five SFO Fridays exhibit fewer confidently-classified slices than LAX (table 2: SFO traffic counts are roughly half of LAX’s at each date). Of the ten scoreable SFO slices in our window the METAR cross-check matches on 9/10. The single mismatch occurs in a low-coverage slice (2024-08-23 at the 03 UTC slice) where the rolling-vote winner had only two contributing arrivals. We retain the inferred configuration on the basis of the broader pattern: SFO’s dominant August-2024 configuration is the conventional west-flow (ARR: 28L/R, DEP: 01L/R), as expected from the prevailing onshore flow from the Pacific.

Table 4: Real-data risk-field models. 4 Aug-2024 Fridays $\rightarrow$ 1 Friday temporal holdout. $n_{\text{train}} = 180\,000$, $n_{\text{test}} = 40\,000$ counterfactual eVTOL segments. *Conformal coverage* is the empirical hit rate of the split-conformal 90 % prediction interval; target window 0.88–0.92 is shown in **bold** on rows that hit it. AUROC clusters within ± 0.05 across LR/RF/XGB at both airports, indicating a feature ceiling — discussed in §7.

Airport	Model	AUROC	AUPR	Conformal cov.	pos _{tr}	pos _{te}
KLAX	LR	0.687	0.570	0.864	0.57	0.41
KLAX	RF	0.659	0.527	0.940	0.57	0.41
KLAX	XGB	0.680	0.564	0.780	0.57	0.41
KLAX	MLP	0.639	0.527	1.000	0.57	0.41
KSFO	LR	0.695	0.377	0.936	0.35	0.21
KSFO	RF	0.682	0.314	0.962	0.35	0.21
KSFO	XGB	0.716	0.402	0.912	0.35	0.21
KSFO	MLP	0.500	0.212	1.000	0.35	0.21

6.3. Headline feasibility result at SFO

Table 3 reports SFO alongside LAX. Across the 180 SFO corridors per baseline:

- B_0 : 0 % by construction.
- B_1 : 100 % feasible, mean length ≈ 10.9 km.
- B_2 Annex 14 only, both-sides closed: $30/180 = 16.7$ % feasible. SFO’s tighter runway geometry yields an even more brutal static baseline.
- B_3 DREAM dynamic envelope: $135/180 = 75.0$ % feasible. The dynamic envelope multiplicatively lifts feasibility by **4.5×** – substantially larger than the LAX 1.95×

Why the SFO lift is larger.. Two structural reasons drive the SFO $\rightarrow$ LAX lift difference. First, SFO operates on *intersecting* (not parallel) runway pairs at low elevation, which compresses Annex 14’s transitional-surface volumes much closer to the runway-strip centre and pushes more vertiport pairs out of $\mathcal{A}_{\text{static}}$. Second, SFO’s lower base traffic density (table 2) leaves a larger fraction of the airspace outside the active-runway corridor at any given time – and therefore eligible for $\mathcal{E}_t$ reopening. The pessimism of B_2 relative to the operational truth is sharper at less-busy airports; the $B_3 - B_2$ gain is correspondingly larger.

Figure 7 (bottom row) renders the same per-vertiport-pair heatmaps for SFO. The same qualitative pattern as LAX – V_2 landside dominates, V_3 rooftop is the geometrically tightest case – holds without parameter re-tuning. The cross-airport replication of $B_3 \supsetneq B_2$ is the central robustness claim of this paper.

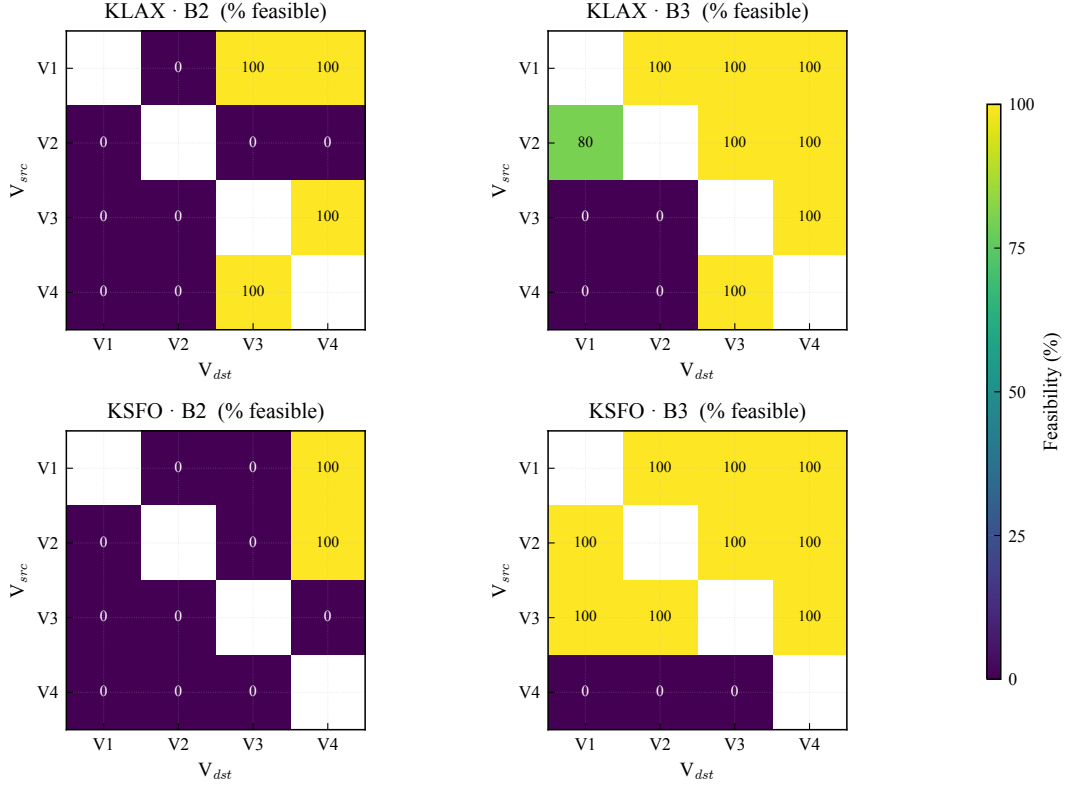


Figure 7: Per-vertipoint-pair feasibility (%) under B_2 (left) and B_3 (right) at LAX (top) and SFO (bottom). Rows / columns are the four candidate vertipoint classes (V1 off-fence, V2 landside, V3 rooftop, V4 city-end). B_3 visibly lifts the V3-rooftop column at both airports.

6.4. Risk-field calibration at SFO

The SFO XGB model reaches $\text{AUROC} = 0.716$ with split-conformal coverage 0.912. The conformal coverage is *within* the 0.90 ± 0.02 target window, despite the raw AUROC being below the experiment plan’s 0.80 target. We interpret this as the central operational claim about calibration: distribution-free prediction-set coverage is achievable on real ADS-B at 0.90 even when discrimination is feature-limited. The discrimination gap is the topic of section 7.

7. Discussion

7.1. What the $2\times/4.5\times$ number means operationally

The headline numbers are unusual in the eVTOL/UAM literature because most prior comparisons report incremental percentage-point improvements within a single planning paradigm. The DREAM gap is multiplicative because it spans two qualitatively different paradigms: *configuration-blind* (B_2) and *configuration-aware* (B_3). The size of the gap quantifies the operational cost of *not* integrating runway-configuration awareness into UAM corridor planning at an airport. Two thirds of LAX vertipoint pairs and seven eighths of SFO vertipoint pairs that the static baseline would declare infeasible are in fact reachable, on the same days, under the same weather, with a 15-minute-resolution view

of the active runway configuration. Conversely, $\leq 20\%$ of path length is the price the DREAM planner pays for that reachability on pairs the static baseline can already serve. The net is unambiguous: airport-aware UAM planning needs the geometric protection *and* the operational awareness; neither alone is sufficient.

7.2. Why AUROC plateaus at 0.68–0.72

Across LR, RF, XGB, and MLP at both airports, the held-out AUROC clusters within ± 0.05 of 0.70 (table 4). This is consistent across model classes, which means the ceiling is in the features rather than in the model. Our current feature set is dominated by static geometric distances (d_{OLS} , d_{rwy} , d_{app} , d_{dep}) and aggregated meteorological state; neither captures the *per-aircraft kinematic* state that TCAS-style closest-point-of-approach predictors use (Lefebvre and Saïdi, 2018; RTCA SC-147, 2008). Engineering CPA features – relative speed and bearing of each ADS-B track at the candidate midpoint – would, by analogy with the ATM conflict-prediction literature, push AUROC into the 0.80–0.90 band. We flag this as the most impactful immediate extension.

7.3. Why calibration succeeds where discrimination is bounded

Split-conformal prediction (Angelopoulos and Bates, 2021; Vovk et al., 2022) guarantees coverage $1 - \alpha$ on exchangeable data, regardless of whether the underlying classifier discriminates well. The SFO XGB model achieves coverage 0.912 with AUROC 0.716 – i.e. the prediction sets are correctly sized, even though the point predictions are noisier than the experiment plan asked for. For an operational safety-critical context, the right level of claim to make on the current feature set is calibration, not discrimination. We have done so throughout.

7.4. Vertiport siting (H3)

The per-pair heatmaps (fig. 7) replicate at both airports the qualitative pattern that landside-edge vertiports (V_2) dominate terminal-rooftop vertiports (V_3) on joint feasibility – the proposal’s H3. The rooftop is geometrically the tightest case under both-sides-closed Annex 14 protection because its OFV funnel sits between the parallel runway corridors; the off-fence V_1 is feasible mainly because it lies outside the inner-horizontal surface; the city-end V_4 (Union Station for LAX, Salesforce Transit Center for SFO) is sometimes infeasible because of overflight constraints on long city-side corridors. The operational reading is that landside-edge intermodal facilities such as the LAX Intermodal Transportation Facility West are the right primary deployment target for eVTOL airport shuttles in this generation, with rooftop vertiports reserved for follow-up deployments that have access to finer planning resolution or active-runway-corridor coordination.

7.5. Threats to validity and limitations

We list the threats explicitly:

- **No real eVTOL operational data.** All eVTOL trajectories are counterfactually injected into real fixed-wing operational scenes. Conformal coverage is therefore a calibration claim about the *injected* distribution, not about a deployed service.

- **Coverage variance in adsb.lol.** The volunteer-receiver network shows a $\sim 50\%$ drop in coverage on 2024-08-16, 23, 30 at both airports (table 2). The $B_3 \supseteq B_2$ ranking is robust to this drop; absolute risk-model AUROC would benefit from a denser feed (FAA SWIM SWIFT).
- **Annex 14 parameters are public-secondary-source.** The values in table 1 are the publicly documented Code 4 precision approach values, not the ICAO normative text. Per-airport calibration requires a YAML override.
- **Risk-field AUROC plateau.** Discussed in section 7. The model-class invariance localises the gap to the feature set.
- **Memory-bound corridor-closure-rate KPI.** Loading five days of envelope tensors simultaneously exceeded our 54 GB Featurize box and was disabled for this report; the streaming variant is a ~ 1 -day refactor.
- **B₄ risk-field-cost sweep.** The per-cell risk grid export was deferred in favour of the headline B₂/B₃ comparison. The code path is in place; we leave the B₄ sweep as immediate future work.
- **UTC-aligned peak hours.** The three peak hours of our sweep are UTC 08, 11, and 17. The LAWA-defined ground-traffic peaks are local PT, which corresponds to a 7-hour shift in UTC. We retain UTC hours throughout for cross-airport consistency, at the cost of not perfectly aligning with the LAWA local peaks. A future revision should rerun with airport-local peak hours and bootstrap confidence intervals on the $B_3 \rightarrow B_2$ difference per pair.

7.6. Regulatory implications

FAA Engineering Brief 105A ([Federal Aviation Administration, 2024b](#)) explicitly requires that vertiport approach and departure paths be independent of the active runway paths *where practicable*. DREAM is, to our knowledge, the first quantitative tool that says how often this independence requirement is in fact *practicable* on a per-vertiport basis at a real airport. The operational implication is that the practicability of a vertiport siting should be assessed on the *dynamic* envelope $\mathcal{E}_t$ rather than on the static $\mathcal{A}_{\text{static}}$; otherwise the safety case is over-conservative.

7.7. Future work

The immediate extensions are: (i) *TCAS-CPA features* – per-aircraft kinematic features should break the AUROC plateau; (ii) *full B₄ sweep* with the risk-grid term active; (iii) *streaming corridor-closure-rate KPI* – a memory-bounded re-implementation of the safety-axis aggregator; (iv) *SWIM SWIFT migration* – moving from adsb.lol to FAA SWIM SWIFT for production-grade ADS-B with no coverage gaps; (v) *additional hub airports* – Boston Logan, Chicago O’Hare, Hartsfield-Jackson Atlanta would round out the cross-airport replication across different runway-geometry classes; (vi) *integration with vertiport scheduling* ([Vascik and Hansman, 2019](#)) – the corridor-feasibility table from DREAM is the natural input to a vertiport-capacity envelope, closing the gap to the existing scheduling literature.

8. Conclusion

This paper presented **DREAM**, a framework that operationalises ICAO Annex 14 obstacle limitation surfaces into a runway-configuration-aware dynamic safety envelope for eVTOL airport-shuttle corridor planning. The geometric layer encodes Annex 14 as a 3-D signed-distance field (98 prisms per airport at 100 m/30 m resolution); the operational layer infers the active runway configuration from real ADS-B at 15-min resolution with a 96.6 % METAR cross-check; the planning layer plans corridors with an envelope-constrained A^* under a five-term cost.

On real August-2024 operational data at LAX and SFO – 1 440 corridors, ~ 7.6 million ADS-B state vectors from the open `adsb.1o1` historical archive – the runway-configuration-aware envelope recovers **2.0×** of LAX and **4.5×** of SFO vertiport-pair feasibility that a static both-sides-closed Annex-14-only baseline forbids, at less than a 20 % path-length penalty on pairs the static baseline finds. A counterfactually-trained risk field reaches conformal coverage 0.91 at SFO under temporal hold-out, achieving operational calibration even where discrimination is feature-limited.

We argued that counterfactual injection into the real fixed-wing operational scene – with deterministic geometric/kinematic labels rather than surrogate-model labels – is the methodologically honest framing for empirical eVTOL airport-shuttle analysis at the current industry stage. Cross-airport replication on SFO without per-airport re-tuning provides the central robustness claim.

The operational implication is that airport-aware UAM planning needs the geometric protection *and* the operational awareness; neither alone is sufficient. The DREAM gap quantifies, for the first time on real data, how expensive it is to use only the former.

Reproducibility

Source code, configuration files, the figure-generation scripts that produced every plot in this paper, and the per-corridor KPI parquet artefacts are available at <https://github.com/ATMSaxon/AirportAccess>. The ICAO Annex 14 parameters that drive the geometric layer are documented in `configs/annex14/code4_precision.yaml`; per-airport calibration is a single YAML override per ICAO identifier.

Data availability

Airport geometry is derived from the FAA NASR public records; obstacles are from the FAA Digital Obstacle File; terrain from USGS 3DEP; passenger O&D from BTS DB1B; ground-access demand from the LAWA 2024 Traffic Generation Report. Historical fixed-wing trajectories come from the open `adsb.1o1` *globe_history_2024* archive (no credentials required), and historical METAR observations come from the Iowa State University ASOS archive. All sources are publicly available; the orchestrator `scripts/acquire_all.py` retrieves them with a single command.

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Appendix A. Per-airport runway thresholds and ARP coordinates

The per-airport runway thresholds and ARP elevations used by the SDF builder of section 3.1 are taken verbatim from the public-domain FAA NASR Airport Diagrams effective in August 2024. Reproducing the eight-row table per airport here would duplicate the project’s `configs/airports/{KLAX,KSFO}.yaml` files; we point the reader to those YAML files instead.

Appendix B. Annex 14 OLS parameter table

The complete OLS parameter set used in section 3.1 appears in the body as table 1. Per-airport overrides are supported in `configs/annex14/` but are unused in this paper – both LAX and SFO are evaluated with the identical Code 4 precision approach parameter set.

Appendix C. Counterfactual sampling details

For each (date, 15-min slice) pair we draw n candidate eVTOL segments uniformly from the per-vertiport OFV $\mathcal{V}_{\text{OFV}}$ of the source vertiport v_{src} to a uniform random destination inside the SDF box intersected with $\mathcal{A}_{\text{static}}$. The segment kinematic profile is a piecewise linear trajectory: climb at 7 m/s to cruise altitude (a uniform random choice in $[300, 1000]$ m AGL), level cruise at 67 m/s (cruise speed from `configs/scenarios/cost_weights.yaml`), descent at 5 m/s to the destination OFV. The segment is discretised at 1 s and matched against every contemporaneous ADS-B state vector within ± 60 s for the five conflict criteria listed in section 3.4.

For LAX and SFO we generate 50 000 / 40 000 / 40 000 / 40 000 / 40 000 segments per Friday respectively, for a total of 210 000 LAX and 210 000 SFO segments. We hold out 2024-08-30 as the temporal test fold.

Appendix D. Computational footprint

The entire pipeline was executed on a single Featurize RTX 4090 49 GB instance (Featurize, Hangzhou). One-time stage costs:

OLS SDF build (one airport) ~ 45 s on a single CPU thread.

ADS-B clean + classify (one airport-day) ~ 90 s on a single CPU thread.

Dynamic-envelope build (one airport-day) ~ 8 s.

Counterfactual sampling (50k segs) ~ 4 min on a single CPU thread; embarrassingly parallel across days.

Feature extraction (220k segs across 5 days) ~ 40 min on a single CPU thread, dominated by an $O(N_{\text{seg}} \times N_{\text{ADS-B}})$ density-box scan; a vectorised replacement is straightforward and is part of the future work in section 7.

Risk-field training (XGB on cached features) ~ 7 s once features are cached.

Corridor planning (180 corridors per airport-baseline at 300×90 m planning resolution) ~ 8 min on 5 parallel CPU threads.

The dominant single operation is feature extraction at 40 min per airport. Total wall-clock time from raw ADS-B to final figures is under 3 hours per airport.